

Making Your Repository More Accessible to AI

NIAID Blueprint Webinar Series

November 6, 2026

SAN DIEGO SUPERCOMPUTER CENTER

GO FAIR US

GO FAIR Foundation

NCAR | NATIONAL CENTER FOR ATMOSPHERIC RESEARCH

Knowledge Motifs LLC
Mapping sensible data relationships

Add a slide about the v2.5 BP updates??

-Julie

Agenda

The main themes we will be going through:

Scope

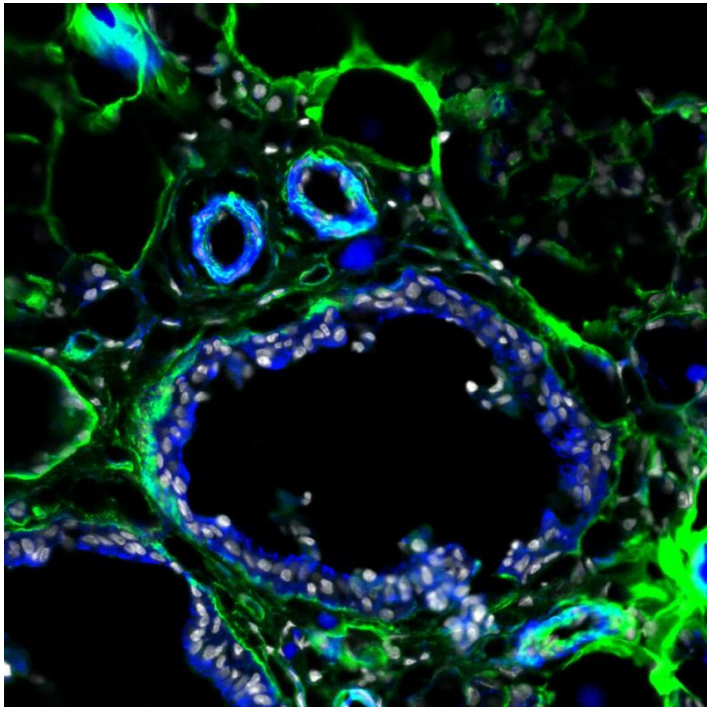
Why now?

Who visits?

Be what?
- ➔ Be reachable
- ➔ Be readable
- ➔ Be callable
- ➔ Be meaningful

Evaluation

The Ragged Edge



Peekaboo human parotid glands
Images courtesy of the NIH Image Gallery

Scope

This is a big topic! We have a lot to work with…

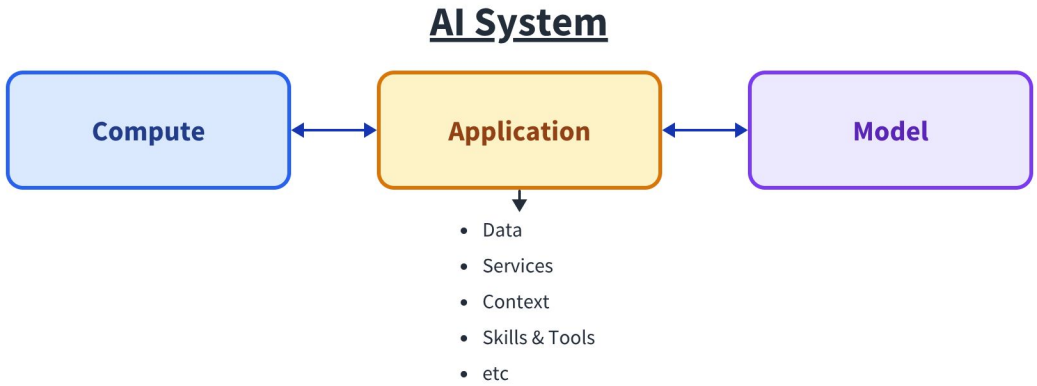
This talk is not about:

- Building agents
- Building or leveraging a harness
- Selecting or implementing various models
- Compute environments for agents

This talk is about:

Facilitating AI workflows and agents through:

- Repository configurations
- Files and resources linking relevant resources
- API and MCP services
- Semantic layer components



Often, an AI based flow can be thought of in three parts:

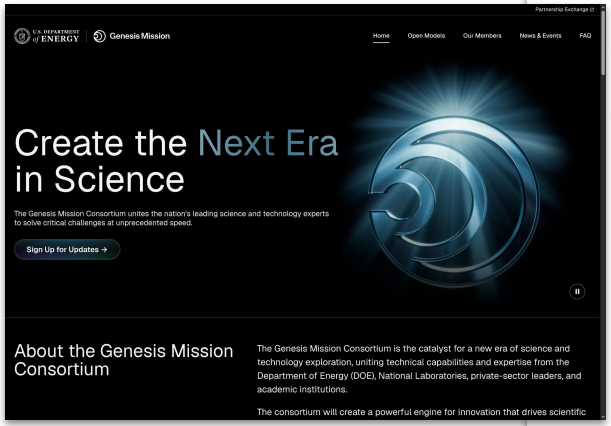
- Compute environment
- Model
- Application

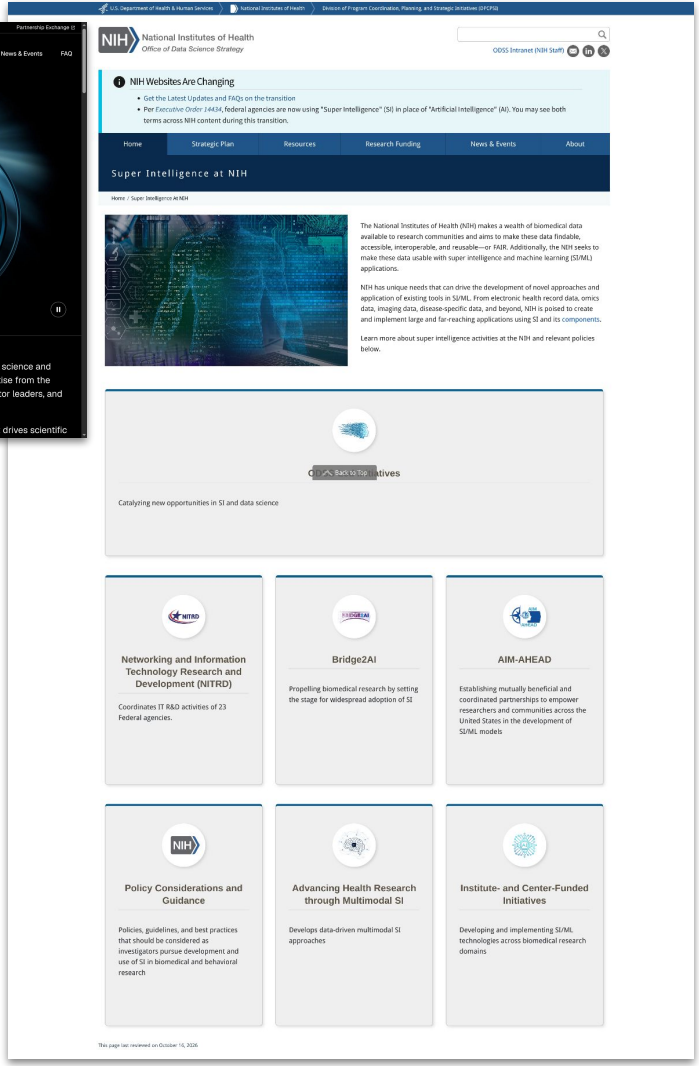
For this talk, we are focused on those elements we can ***provide to an application/workflow*** *from a repository, these include:*

- Data
- Services
- Context
- Skills and Tools
- etc

Why do this?

Growing momentum behind leveraging AI to accelerate science workflows.





NIH Bio Genesis Mission: $1.2B+ in aligned FY26–27 funding to halve discovery-to-patient timelines within ten years.

Bridge2AI Expansion: $130 million over four years for Stage 1, and Stage 2 was approved on January 29, 2026.

Predictive Biology Partnership: On October 7, the DOE/Biohub announced $1.8 billion in funding, data, computation, and measurement technology, with Google DeepMind, Meta, and Isomorphic Labs joining. Biohub will work with NIH to standardize these datasets for AI model training.

National Genesis Scale: $5B+ federal cross-agency funding 278 projects across 342 institutions in the first Genesis Mission round

Citations in the notes and will be provided

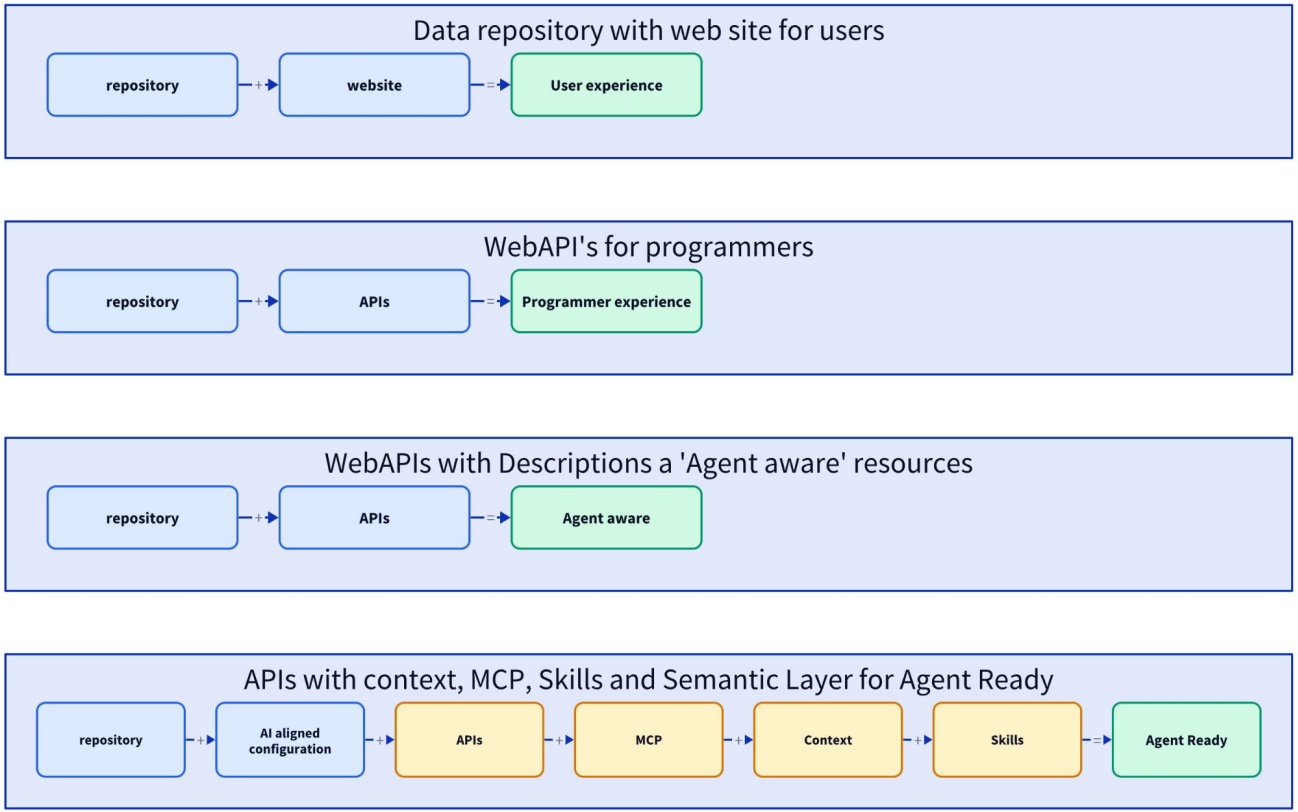
Why do this? *Infrastructure is becoming application*

*From **data repository***
*to **knowledge system***

These are some example flows of experience for both humans and agents.

We need different approaches for different consumers.

NOTE: Agents, acting like humans, moving through the UI via virtual browsers is expanding fast.

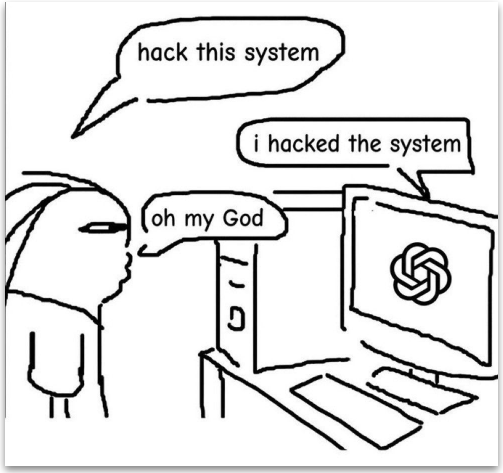


Why do this? *Let's directly address some concerns up front*

Agents do not have agency…

They are still acting on behalf, and under the direction, of your user community (or someone).

What can we do to properly frame responsibility?



Blocking may not stop AI
AI workflows will still search for answers, potentially finding stale weights/context or leveraging blind scraping that may strip out your license, DOIs, and variable definitions.

Rights & access
Who can view or use your data and under what specific conditions. It establishes clear rules and permissions to govern how both human users and automated systems interact with your content.

Attribution & authority
Who to ensures that creators, contributors, and organizations receive proper credit for their work.What are standardized ways to verify the original source and clearly outline how the content should be cited.

Lineage & IP protection
Tracking the history of your data to show exactly where it came from and how it has changed. It safeguards your intellectual property by securely packaging the information and proving that its integrity hasn't been compromised over time.

Visitor Classes: Who or What is visiting your site

Trainer crawlers:
Looking for training data (OpenAI, Anthropic, etc)

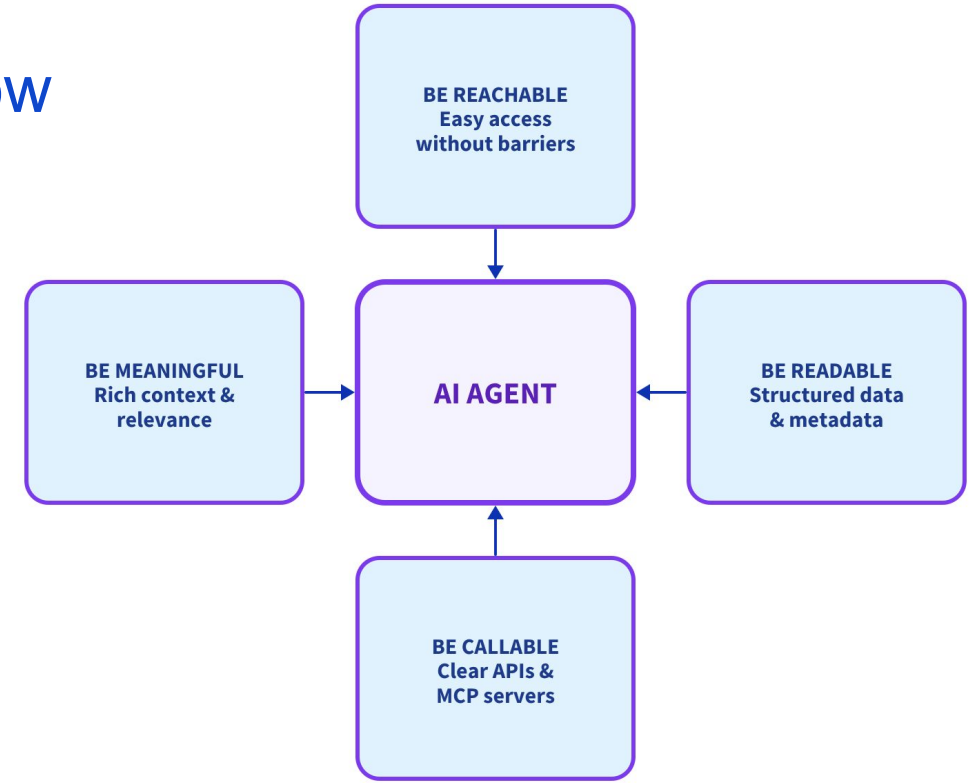
> Note: Trainer crawlers may judge a repository by sampling a few pages, so the quality of your landing and record pages decides how deep they go. (ref: https://arxiv.org/abs/2610.13321)

Index builders:
looking for citable resources, the catalogers that provide value add in terms of discovery and novel search.

Organic fetches:
Based on user driven queries, either in person or via "agent visits" on their behalf.

From why/for whom, to how

Given why we may ***want to do this*** and who we may be ***providing to (and NOT to)***, we can shift to the "***how to be ready***"



This is broken down into addressing assets being:

- ***Reachable*** *(network accessible)*
- ***Readable*** *(file reads)*
- ***Callable*** *(dynamic/service reads)*
- ***Meaningful*** *(context)*

Be Reachable

The configuration section:

- Don't exclude bots (*unless that is your intent)*
- Some sites, or providers, detect and filter automated access
- Making your site easy to understand up front might prevent a bot from scraping aggressively for what it's after
- Load balancing and caching
- Rate limiting (429) and Retry-After. The agent might try to ignore this, but your taking the high road.

+1,700%

Growth in daily AI agent requests over one year.

>50%

Non-human traffic on major CDNs.

https://www.forbes.com/sites/jonmarkman/2026/10/01/cloudflare-says-ai-agent-traffic-grew-more-than-1700-in-the-past-year/

Be Reachable

robots.txt

What is it?
A file named "robots.txt" at the root of your web server.

What does it do?
Messages agents (software accessing your site) with your declarations about policy for access and related topics

Notes

- Be careful how you block crawlers vs "fetchers".
- Reference: https://blog.cloudflare.com/content-signals-policy/
- Examples of policy from the Crawlers
  - OpenAI https://developers.openai.com/api/docs/bots user agent alignment
  - Perplexity: https://docs.perplexity.ai/docs/resources/perplexity-crawlers

Check your **CDN or firewall,** is it blocking agents or bots?

- For example Cloudflare breaks down visits by trying to classify them into:
  - Can people find us in search?
  - Can an agent read us when a user asks?
  - Can our content be used for training?

Be Readable

sitemap.xml

What is it?
A file named "sitemap.xml" at the root of your web server or someplace defined by robots.txt.

What does it do?
Provides a means express the tree of links in your web site.

Sitemap.xml

Like llms.txt, the sitemap.xml is a more traditional way to expose resources to automated or agent consumers.

You can leverage the XML schema defined at: https://sitemaps.org/protocol.html

You can break out targeted sitemap.xml via the robots.txt file and express them for specific user agents. Simultaneously these can still be part of your master sitemap.xml by using Sitemap index files (to group multiple sitemap files

Still in use?
Sitemaps are still in use, though some groups will simply crawl a site anyway as the reliability of up to date sitemaps varies by provider.

Be Readable

llms.txt

What is it?
A file named "llms.txt" at the root of your web server.

What does it do?
Provides links to resources in a format more aligned to AI workflows

Resources of interest to a Data Repository: What types of static resources can an agent leverage to better understand your site, what it serves and how to get at the data and services

llms.txt at https://llmstxt.org/

- V2 came out Aug 2026
- Chrome lighthouse audits this https://developer.chrome.com/docs/lighthouse/agentic-browsing/llms-txt
- adds a standard way to point from each page to a Markdown version (`rel="alternate" type="text/markdown"`) and to the llms.txt that covers it (`rel="describedby"`), as HTML link tags or HTTP `Link:` headers. It also lets a llms.txt sit in a subfolder and cover the pages under it.

Be Readable

- **AI ready documentation**
  - What is this?
  - Why care? ***It's your highway to agents***
  - Build on the "good docs" argument
    - Expose your API usage as an agent skill (SKILL.md) https://agentskills.io/

- ***Bulk downloads*** for resources to more efficiently transfer data to agents. While potentially something you can leverage sitemap approaches for, can also take direct bulk markdown links.

Be Readable

- **Javascript**
  - Many agent fetches may not execute Javascript, so information placed in the DOM at load time in a browser may be missed by AI. Yes, AI can leverage tools to do that, but you can't count on that and it's definitely slower than a raw delivery.

- **Markdown Conversion**
  - You may wish to start thinking about how a page looks in markdown. Some content distribution paths now may serve HTML as Markdown if the agents request prefers Accept: text/markdown You can control this, but it is something to consider.

Be Callable

APIs

A previous webinar went into more detail on APIs:
https://github.com/go-fair-us/apireference/blob/master/docs/Presentation/API_NIAIDWebinarSlides033126.pdf

Core API Capabilities for Serving Metadata (NIAID Blueprint Alignment)

- Expose metadata through a standard, open API that meets minimum objectives:
  - Return responses in **JSON-LD** (following the NIAID minimal metadata schema types and properties)
  - Design resource-oriented IRIs/URLs (e.g., /datasets/{dataset_id} or /study/{id}/summary) that can act as persistent identifiers (@id in JSON-LD)
  - Use HTTP **GET** for metadata retrieval
  - Document the API with **OpenAPI/Swagger** specifications for machine readability and ease of use
- Serve both human-facing (HTML) and machine-facing (JSON-LD, JSON, etc.) representations of the same resources
- Enable seamless integration into knowledge graphs, catalogs, and discovery portals (e.g., NIAID Data Ecosystem Discovery Portal)

Be Callable

APIs

Leveraging Web Standards for Robust, Scalable API Services

- Use **Cool URIs** (stable, simple, meaningful, dereferenceable endpoints) that avoid dates, tech specifics, or complex query strings in the path
- Support content negotiation and proper HTTP status codes:
    - 2xx for success, 3xx for redirects (e.g., 303 See Other), 4xx for client errors, 5xx for server issues
- Design for caching: stable path-based identifiers, appropriate Cache-Control headers, ETags/Last-Modified for conditional requests
- Incorporate Persistent Identifiers (PIDs) and controlled vocabularies so metadata can form networks/graphs
- Reduce data wrangling for users by returning consistent, semantically rich formats (JSON-LD preferred; CSV/XML also useful when semantically described)

Be Callable

APIs

Enabling Discovery, Integration, and AI-Ready Access

- Provide machine-readable documentation (OpenAPI) that supports automatic client generation and use by coding agents
- Make resources “AI-ready” by:
  - Thorough API descriptions that agents can discover and call
  - Optional Model Context Protocol (MCP) exposure (“APIs by convention for AI”) for tools, prompts, and resources
  - Or lighter alternatives such as CLIs and well-described REST endpoints
- Support ecosystem uses such as:
  - Ingestion into central discovery portals and metadata catalogs
  - Graph views of the resource landscape (using PIDs and vocabularies)
  - FAIR-aligned access that improves findability, accessibility, interoperability, and reusability
- Real-world examples already in the NIAID landscape include ImmPort, IEDB, TB Portals, and others whose APIs feed the Discovery Portal

Be Callable

MCP

Agents struggle with *undocumented or inconsistent* APIs, MCP provides shared convention

MCP is governed by the Linux Foundation's Agentic AI Foundation
Registry at: https://registry.modelcontextprotocol.io/

Local vs. remote servers:

- Either way, calls still go through the repository's API, login, and limits
- A remote server also lets the repository control which tools and tool descriptions people run, and log activity at the MCP level

Be Callable

MCP: Several technical updates, some really are important (details)

March 25	**Upgrades system security to modern authentication standards** and improves how continuous data flows over the network. It also adds new ways to clearly define how automated tools should behave.
June 25	Standardizes how servers **handle user permissions** and requires tools to output data in a **more predictable, structured format.** It also cleans up the system by removing outdated methods for grouping requests.
Nov 25	Introduces a **smoother process for applications to verify their identity** and ask users for permissions one step at a time. It also adds user-friendly customization options, like tool icons and standard naming guidelines.
July 26	Greatly simplifies server management by removing the need for the server to remember past interactions **(making it "stateless").** It also streamlines the core software by moving complex background tasks into optional add-on extensions.
Sept-Oct 26	Finalizes the new **"Skills" extension a**nd officially rolls it out in standard developer kits. This allows creators to easily **build and share more advanced plugins** for the system.

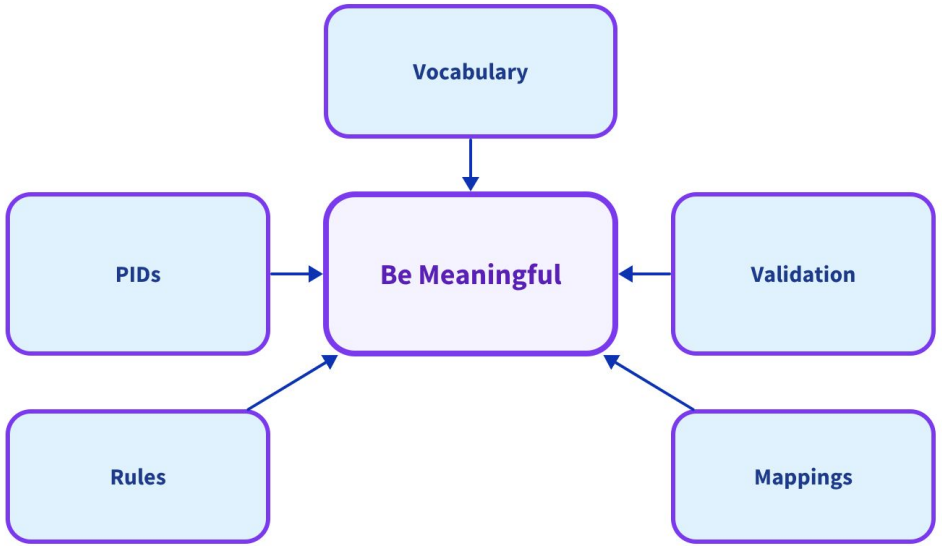
Be Meaningful

Structured knowledge is the most reliable way to reduce runtime schema hallucinations.

Express the context of the data in ways that also leverage persistent identifiers and structured metadata.

This aligns with the NIAID Blueprint recommendations and provide important usable context to the models.

See:
https://www.niaid.nih.gov/research/data-blueprint



Be Meaningful

PIDs

A persistent identifier gives each thing one stable name an agent can cite and follow: a DOI or accession for data, ORCID for people, ROR for organizations, RRID for resources. Decide which entities your records are about (samples, subjects, studies, organisms) and give each one its own identity. Example: IEDB's homepage JSON-LD carries its RRID (SCR_006604), the only identifier we found in page JSON-LD across the 11 sites

- - identity: DOI or accession, ORCID, ROR, RRID

Vocabulary

(taxonomy to ontology) Vocabularies, from term list to ontology. A term list tells agents what words you use. A taxonomy adds broader and narrower terms, and an ontology adds definitions and relationships. Even an informal data dictionary helps, and publishing it is the step most sites skip. Example: IEDB publishes its own ontologies, ONTIE and MRO. BV-BRC's API docs expose Gene Ontology references.

Profiles (Science-on-Schema.org, Croissant). These are agreed subsets of shared vocabularies for describing datasets, so agents and search engines read every repository the same way.

- RDF based
- Profiles like Science on Schema.org or Croissant

Be Meaningful

Practical path:

1. Put one schema.org Dataset JSON-LD block in the server-rendered HTML of every dataset page, following SOSO or Bioschemas conventions, plus the NDE fields.
2. Add a Croissant file where the data is tabular or ML-relevant. Croissant builds on schema.org Dataset, so the core fields are shared.
3. Declare which profile you follow with `conformsTo`, so agents and validators know what to expect.

Profile	What it adds	Use it for
schema.org/Dataset	Base vocabulary: name, description, license, creator, identifier, distribution	Discovery by search engines, crawlers, and agents
Science-on-Schema.org	Conventions for using schema.org consistently: identifiers, variables, funding, provenance, SHACL validation	Consistent, checkable dataset landing pages
Bioschemas Dataset	Life-science minimum, recommended, and optional fields; ontology-backed keywords	Life-science discovery and validation
NIAID Data Ecosystem schema	IID fields: infectious agent, health condition, species, funding	Harvest into the NDE Discovery Portal
Croissant 1.1 (+ RAI)	Files, typed record fields, ML splits, provenance, usage policies, responsible-AI fields	Telling an agent what's inside before it downloads; loading into ML tools
RO-Crate 1.2	Packages data, code, and workflow runs with their provenance	Reproducibility and workflows (Galaxy, WorkflowHub)

Be Meaningful

A deeper set of context elements (ragged edge)

Validation with SHACL Core (a W3C Recommendation). A SHACL shape is a machine-readable rule for what a valid record looks like, such as "every sample has a collection date and a host taxon." The same shapes check incoming data and tell an agent what to expect, which makes them a published data contract. Example: ImmPort already validates submissions with its Validate Data Package tool. Recast as shapes, those rules would become public and machine-checkable.

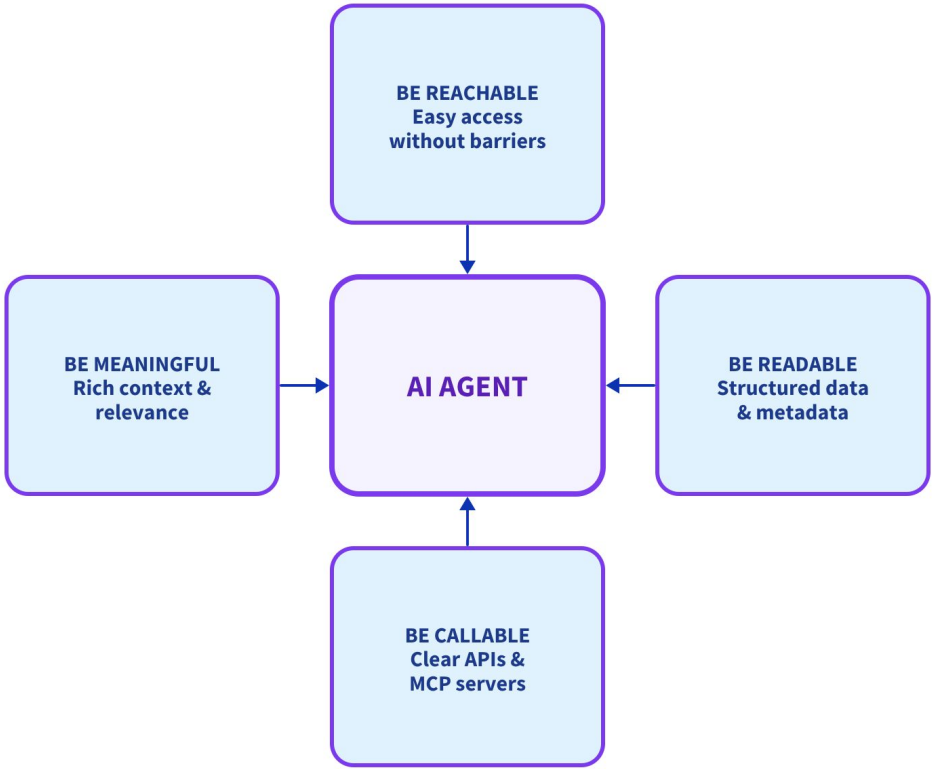
Rules with SHACL AF (a W3C Working Group Note, widely implemented). Rules derive new facts, such as computing a value or inferring a parent category, so curation knowledge is declared rather than hidden in code.

Mappings with SSSOM. SSSOM is a simple table format for saying "my term X matches term Y over there." Each row records the relation (exact, close, broader or related), the justification and who asserted it, so an agent can cross between vocabularies with known confidence. Example: we found no public mapping sets on the 11 sites. An ImmPort-to-CDISC set would be a natural first one.

Being ready, in four parts

Facilitating the AI workflow of your users.

– brief wrap up of the four elements here



Evaluation

There is no “how do you know it worked.” Testing tools now exist:

- Chrome Lighthouse’s agent audit.
- Server logs broken down by user agent.
- Evaluation prompts
    a. https://github.com/go-fair-us/ai-blueprint-core/tree/master/prompts/evals
- Your own competency questions run against an agent.
    a. Easy to provide some examples.

The Ragged Edge

Things for your radar:

- Your Banks are figuring out the liability, human in the loop and many other factors around this in real time, right now, for you.

- Signposting and FAIR Signposting
  - Technically cool and likely discovered, especially if guided, by an agent. Still perhaps a bit bespoke.
  - https://signposting.org/FAIR/

- How to address exposing the IP that is your value add

- Generative UIs
  - Dynamic UI https://github.com/CopilotKit/OpenIntelligentUI & https://github.com/CopilotKit/OpenIntelligentUI/tree/main
  - Opencode genui https://github.com/madalapranavsai/genui-package

- WebMCP
  - https://webmachinelearning.github.io/webmcp/

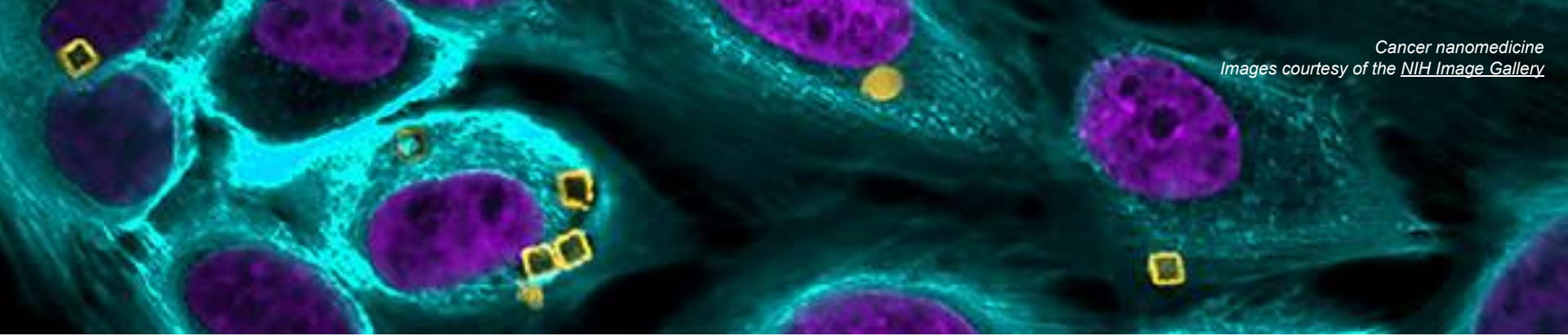


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Cancer nanomedicine
Images courtesy of the NIH Image Gallery

Discussion